



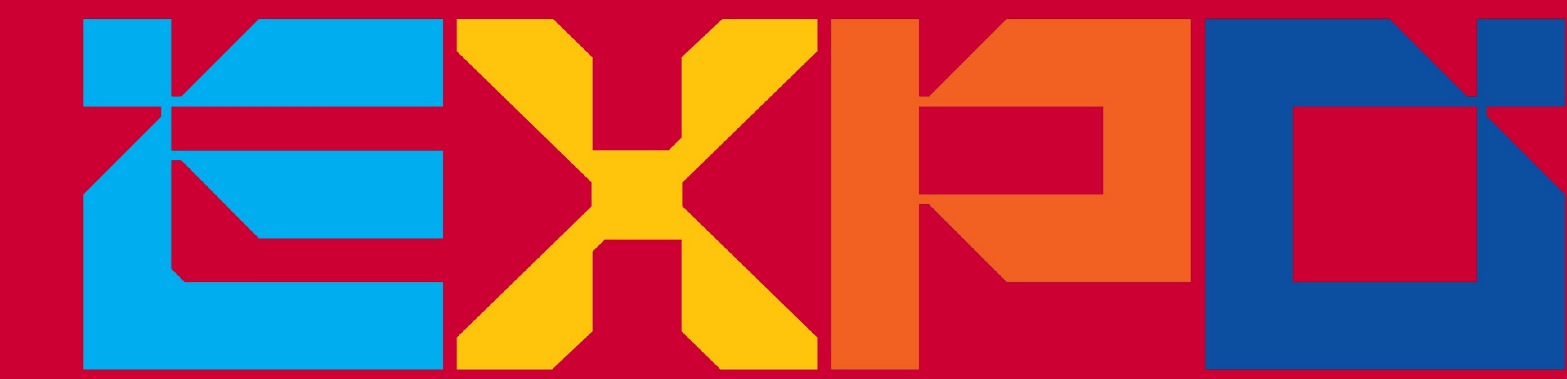
RUTGERS-NEW BRUNSWICK
School of Engineering

RideAware

A Smart Helmet for Cyclists

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Capstone
Senior Design



Rutgers School of Engineering

Abstract

Cyclists often ride alongside vehicles on high-speed roads or in areas without dedicated bike lanes, forcing them to turn their heads to check blind spots. This reduces awareness and can impact stability, while existing solutions lack real-time, helmet-integrated detection. This project introduces RideAware, a smart helmet that detects approaching vehicles and provides intuitive haptic feedback and hands-free signaling. The system is designed to deliver real-time alerts based on the position of nearby vehicles, allowing riders to react quickly without shifting their focus. By integrating these features into a single wearable platform, it improves awareness while maintaining comfort and ease of use, ultimately enhancing overall safety and rider confidence.

Motivations/Objectives

Motivations

- Increasing number of cyclist accidents due to limited rear visibility
- Lack of real time awareness of approaching vehicles
- Unsafe need for riders to turn their heads while in motion
- Existing solutions are limited or not fully integrated

Objectives

- Develop a helmet integrated system for real time vehicle detection
- Provide directional haptic feedback without distracting the rider
- Implement hands free LED turn signaling using head gestures
- Design a lightweight, safe, and wearable prototype

Project Challenges

Real Time Processing Constraints

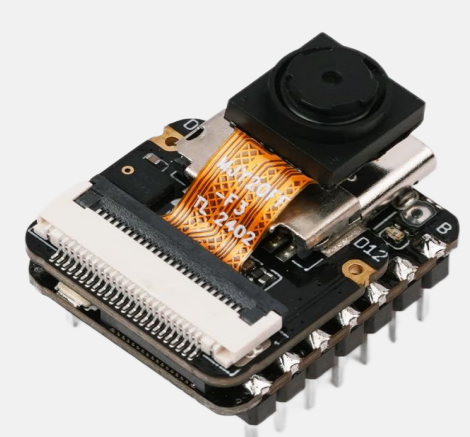
Achieving real time vehicle detection requires balancing model accuracy with processing speed. Running computer vision efficiently on embedded hardware remains a key challenge.

Power Consumption and Management

High performance processing introduces increased power demands, which can be difficult for a wearable system. Developing a reliable and portable power solution while maintaining battery life is an ongoing challenge.

System Reliability and False Positives

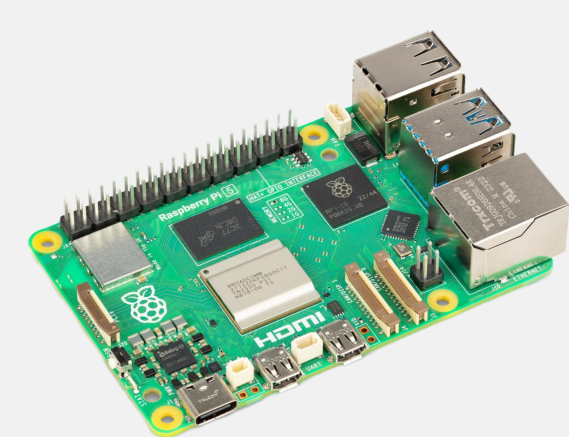
Maintaining consistent detection across varying environments such as lighting and traffic conditions is challenging. Reducing false alerts while preserving sensitivity requires continued tuning and testing.



ESP32-S3



Raspberry Pi Zero 2 W → processes 1-2 fps



Raspberry Pi 5 → processes 20-25 fps

System Design

Vision Hardware

A rear mounted camera continuously captures the FOV behind the cyclist and sends recorded data to the Raspberry Pi 5 to be processed.

Vehicle Detection

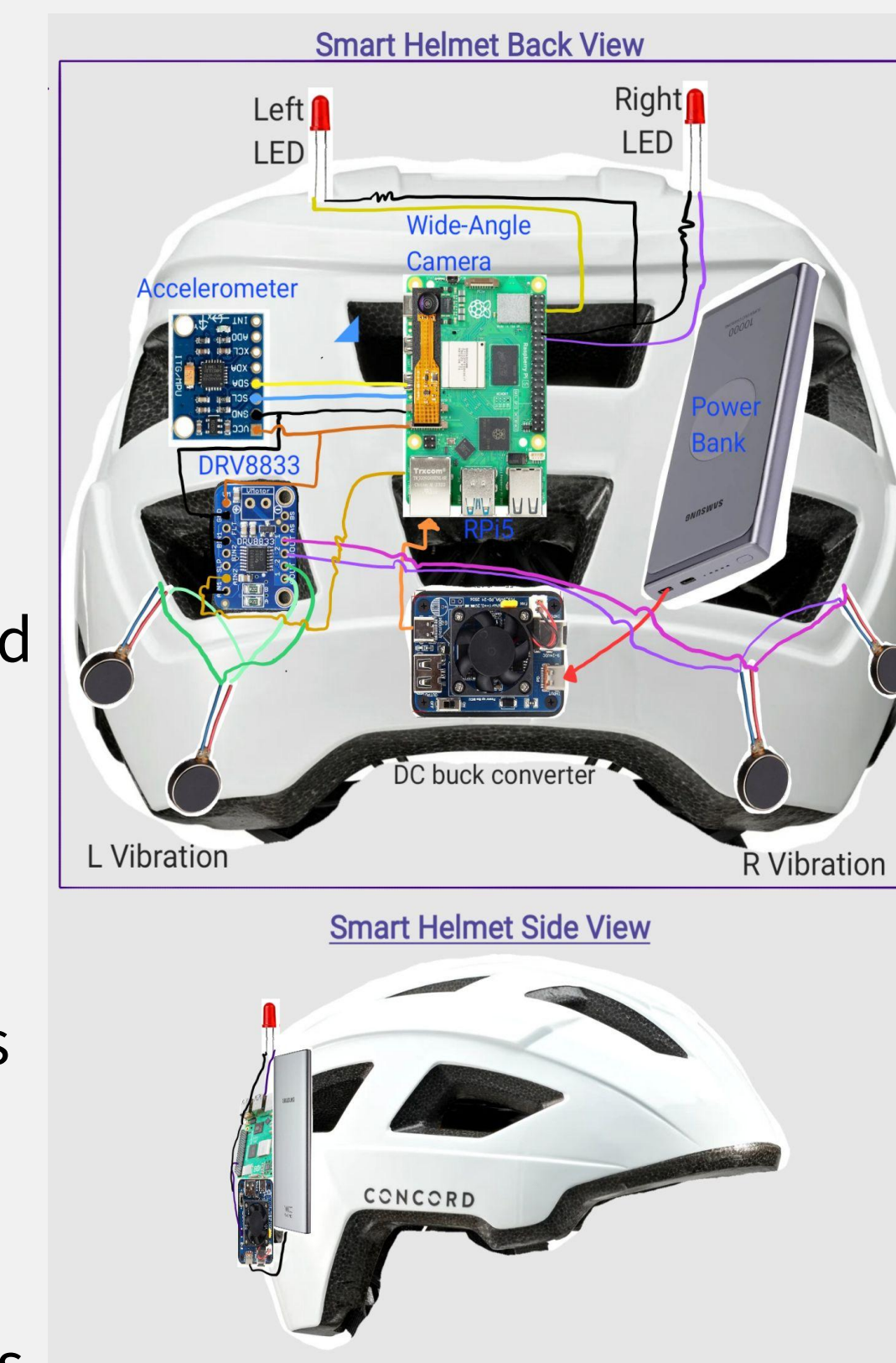
Utilizing the YOLOv4-Tiny lightweight computer vision model, it identifies approaching vehicles and estimates their position relative to the rider.

Haptic Feedback

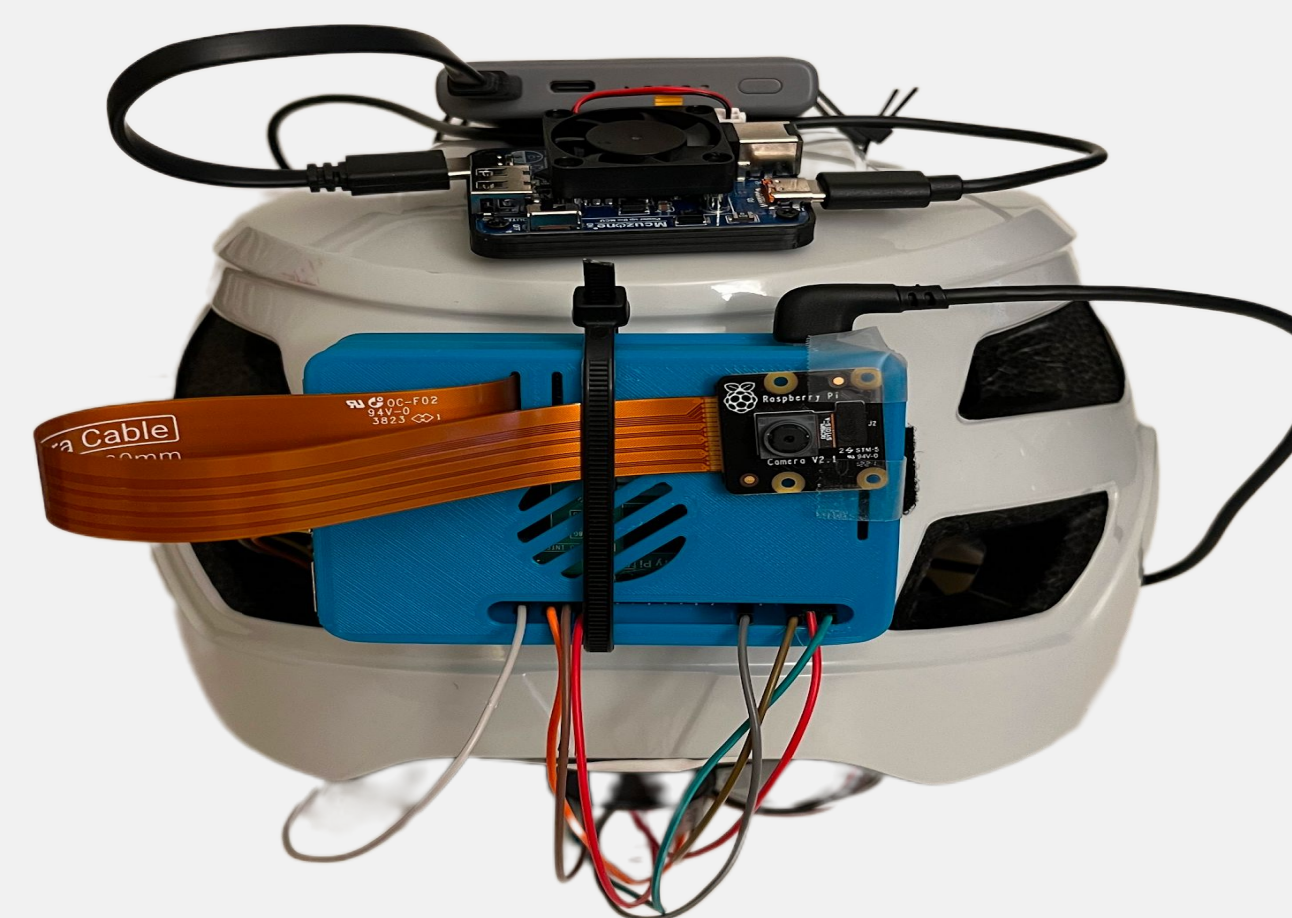
Two DRV8833 motor drivers control the vibration motors on the interior of the helmet. Each side has one motor in the front and one in the rear.

Gesture Based Signaling

Rider can activate or deactivate LED turn indicators with 2 head tilts to left or right side.

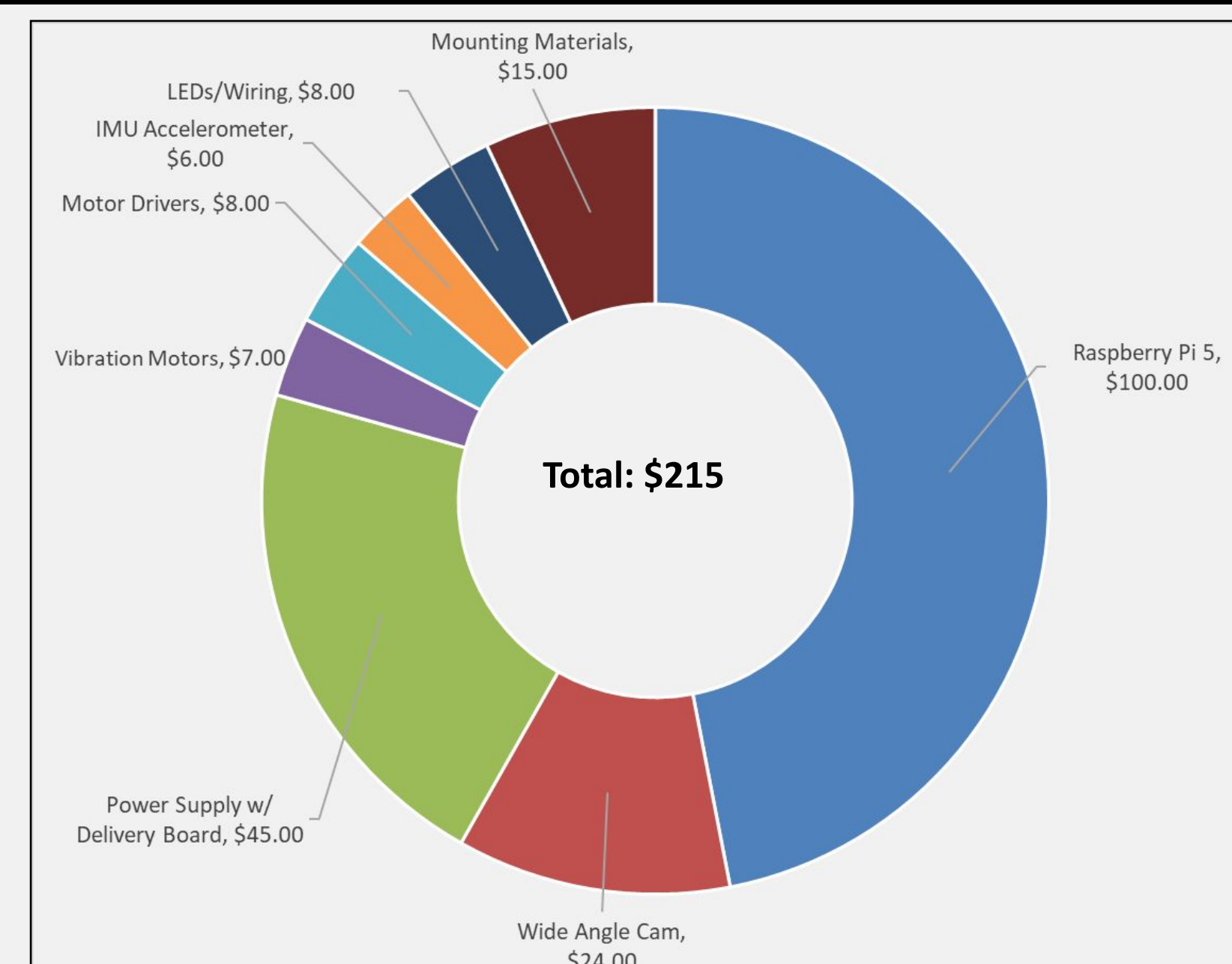


Initial System Design of Helmet



Implemented System Design

Project Cost



The total prototype cost for RideAware is approximately \$200-\$215, primarily driven by processing hardware, sensing components, and power management.

Cost efficiency can be improved through bulk purchasing, custom PCB integration, and optimized component selection. In small-scale production (100-500 units), the estimated cost per unit could be reduced to approximately \$110-\$130. Additional reductions are possible by replacing development boards with dedicated embedded hardware and simplifying the overall system design.

Results

Alert Factors:

- Approaching vehicle area
- Object area changes in between frames
- Relative direction to cyclist
- Distance from rider
- Persistence counter to rule out false positive

```
area=73466, side=CENTER, approaching=False, persist=0
persist_count=1
YOLO ran
area=53694, side=CENTER, approaching=False, persist=1
persist_count=2
YOLO ran
area=53872, side=RIGHT, approaching=False, persist=2
persist_count=3
YOLO ran
area=50962, side=CENTER, approaching=False, persist=3
persist_count=4
YOLO ran
area=52530, side=RIGHT, approaching=False, persist=4
persist_count=5
YOLO ran
area=62408, side=RIGHT, approaching=True, persist=5
persist_count=6
APPROACHING VEHICLE
ALERT LEVEL: DANGER, SIDE: RIGHT
YOLO ran
area=69000, side=RIGHT, approaching=True, persist=6
persist_count=7
ALERT LEVEL: DANGER, SIDE: RIGHT
YOLO ran
area=75660, side=RIGHT, approaching=True, persist=7
persist_count=8
ALERT LEVEL: DANGER, SIDE: RIGHT
```

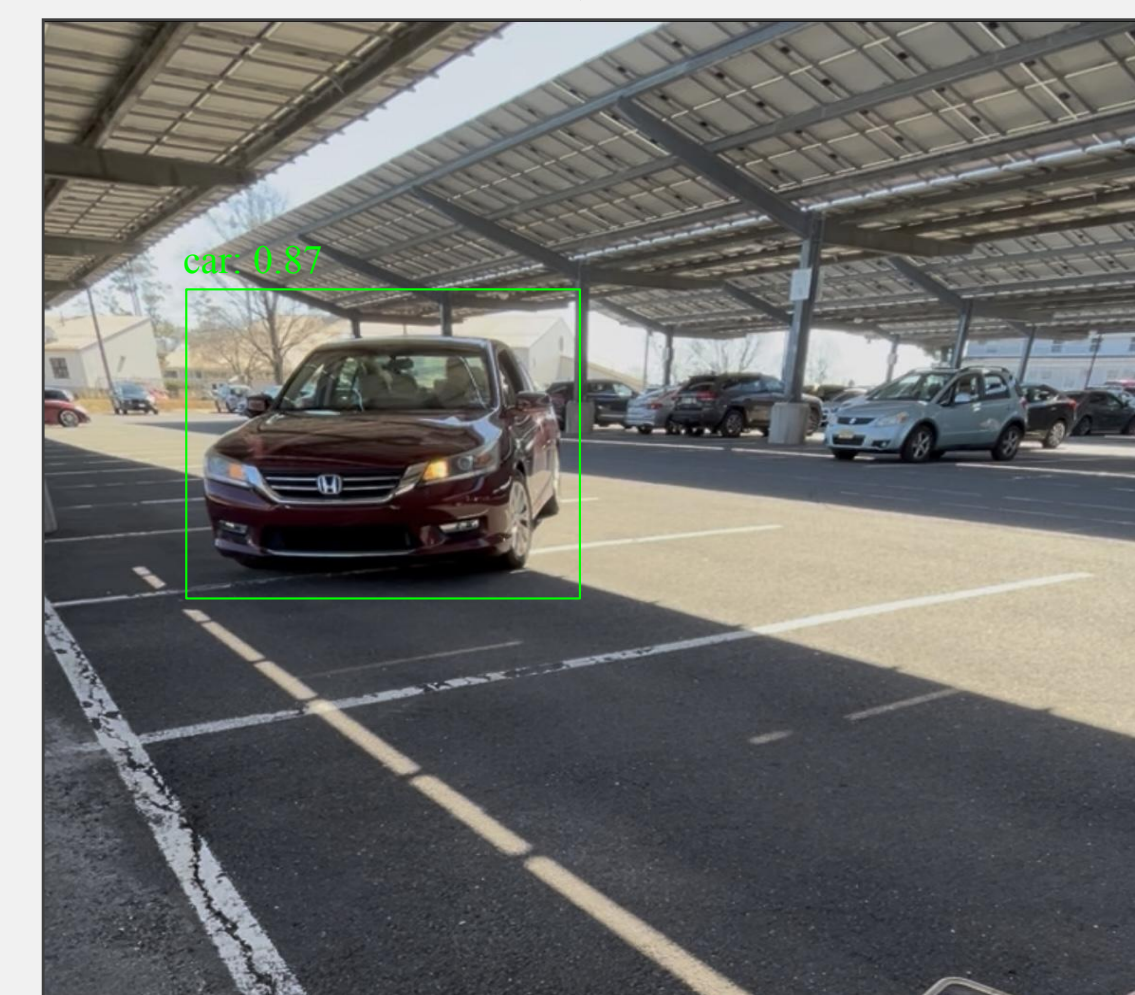
Live Detection Log with Alert Decision

System functions:

- Vibration patterns vary according to vehicle distance and position
- Motion-activated turn signal indication available to warn other motorists



Freeze frame images of a vehicle approaching the system with confidence levels calculated



Project Impact and Future

Our project:

- Improves cyclist safety by increasing awareness with a wearable device
- Combines real-time detection, feedback, and indication into a system
- Useful in a variety of environments and implementation can be tweaked depending on rider's scenarios

Future work includes:

- Increased detection capabilities
- Utilizing a smaller and more task focused SoC instead of the Pi 5
- Significantly lower power consumption
- Custom detection/feedback settings
- Cleaner implementation of components



Single Purpose Chip with Camera for Computer Vision

Acknowledgments

We would like to extend our gratitude to our advisor, Professor Minning Zhu, for his guidance and for continually challenging us to improve our design and approach throughout the project. Additionally we appreciate Kevin Wine's around the clock support throughout this project.